

- VII a) Define Z Transform, Important Theorems of Z Transform and its relation with Laplace Transform. (10)
- b) Solve the second order difference equation using Z transform method (10)
- $Y(k+2) + 0.5Y(k+1) + 0.2Y(k) = u(k)$ where $u(k) = 1$ for $k = 0, 1, 2, \dots$

OR

- VIII a) Determine the Z transform of the following sequences. (10)
- (i) $f(kT) = kT \sin 2kT$
- (ii) $f(k) = \begin{cases} 1k = 0, 2, 4, 6, \dots & \\ -1k = 1, 3, 5, 7, \dots & \end{cases}$
- b) Find the inverse Z transform of the following function (10)
- (i) $F(z) = 10z / (z-1)(z-0.2)$
- (ii) $F(z) = 10z / (z-1)(z^2 + z + 1)$

IX

Write short notes on

- (a) Amplidyne
(b) Pneumatic Controllers
(c) Thermal Systems
(d) DC Servo motor

(20)

OR

X

Write short notes on

- (a) AC Servo motor
(b) Hydraulic Controllers
(c) Stepper Motors
(d) Liquid Level System.

(20)

BTS(C) 055(E)

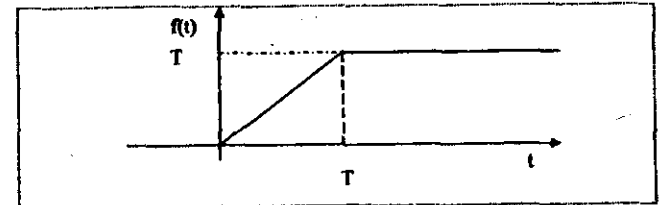
B.Tech. Degree VI Semester Examination
July 2002

ME 605 CONTROL SYSTEMS ENGINEERING
(1999 Admissions)

Time: 3 Hours

Max. Marks: 100

- I a) Distinguish between open loop systems and closed loop systems with examples. Explain the term analogous systems. (10)
- b) State and explain Mason's gain formula. (10)
- OR
- II a) Obtain the Laplace transform of the function $f(t)$ shown in the fig. below. (6)



- b) Find the Inverse Laplace transform for the function. (6)

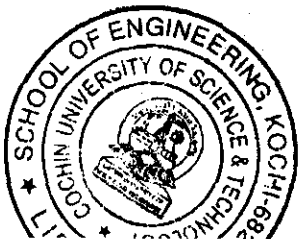
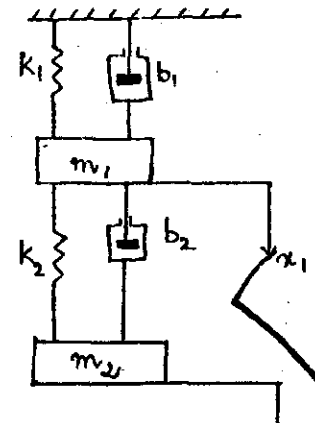
$$F(s) = s + 1/s(s^2 + s + 1)$$

- c) Solve the following differential equation (8)

$$2\ddot{x} + 7\dot{x} + 3x = 0; x(0) = 3; \dot{x}(0) = 0$$

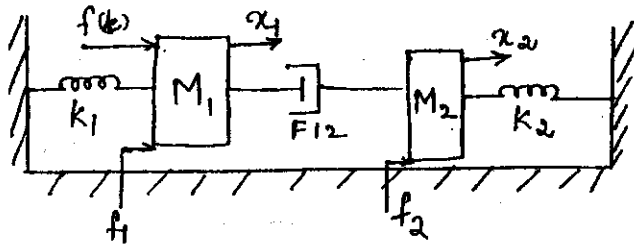
III

- a) Using the force - voltage analogy, obtain an electrical analog of the mechanical system shown in fig. (8)



III

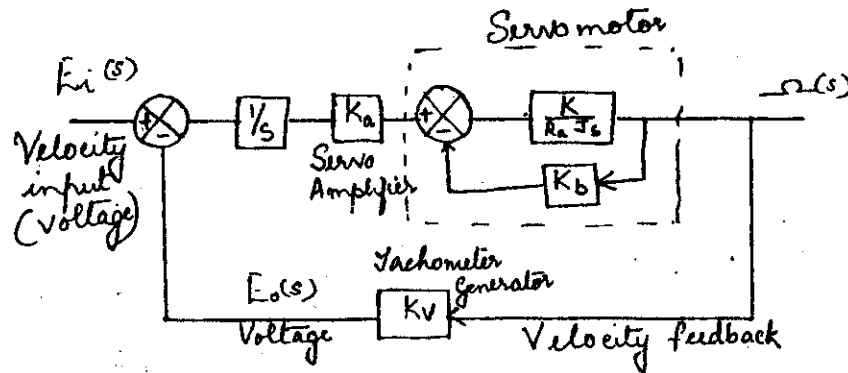
- b) Write the differential equations for the mechanical system shown in fig. Also obtain an analogous electrical circuit based on force-current analogy. (12)



OR

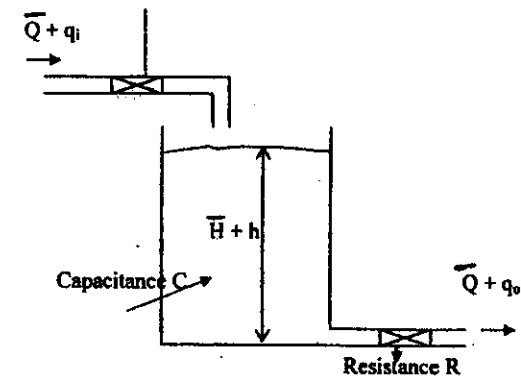
IV

- a) The block diagram of a speed control system shown in fig. Obtain the transfer function between $\Omega(s)$ and $E_i(s)$. (10)



- b) Consider the liquid level system shown in fig. Assuming that $\bar{H} = 3\text{m}$ and $\bar{Q} = 0.02\text{m}^3/\text{sec}$ and the cross sectional area of the tank is equal to 5m^2 , obtain the time constant of the system at the operating point $(\bar{H}, \bar{Q})$. Assume that the flow through the valve is turbulent. (10)

Contd...3



V

- a) The characteristic equation of a servo system is given by

$$a_0 s^4 + a_1 s^3 + a_2 s^2 + a_3 s + a_4 = 0 \quad (12)$$

Determine the condition which must be satisfied by the coefficients of the characteristic equation for the system to be stable.

- b) Write the differential equation for second order transients for a unit step input function and explain the various terms. (8)

OR

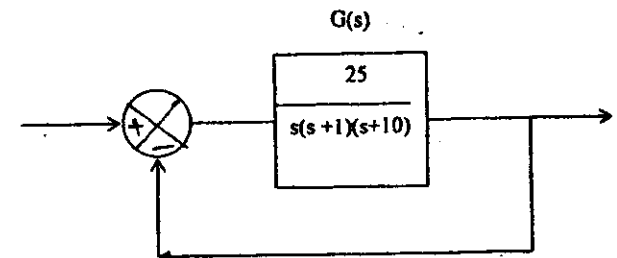
VI

- a) Consider a unity feedback control system with the following open-loop transfer function

$$G(s) = \frac{1}{s^3 + 0.2s^2 + s + 1}$$

Draw a Nyquist plot of $G(s)$ and examine the stability of the system. (10)

- b) Consider the system shown in figure below. Draw a Bode diagram of the open-loop transfer function $G(s)$. Determine the gain and phase margins. (10)



Contd...4